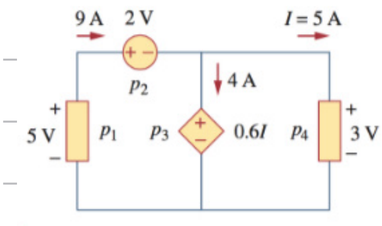
①

② 2.15 3.4

Week 1

P1.7



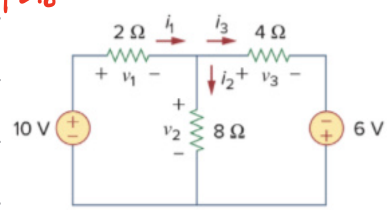
$P_1 = -(4+5)\times 5 = -45W$

$P_2 = 9\times 2 = 18W$

$P_3 = 4\times 3 = 12W$

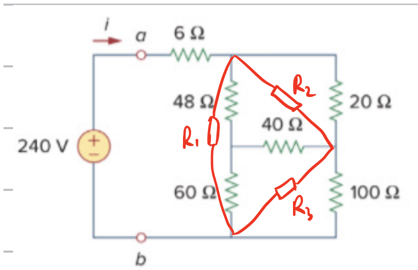
$P_4 = 3\times 5 = 15W$

P2.8



$$\begin{cases} V_1+V_2-10=0 \\ V_3-6-V_2=0 \\ i_1=i_3+i_2 \\ V_1=2i_1 \\ V_2=8i_2 \\ V_3=4i_3 \end{cases} \Rightarrow \begin{cases} i_1=3A \\ i_2=0.5A \\ i_3=2.5A \\ V_1=6V \\ V_2=4V \end{cases}$$

P 2.15



$$R_1 = \frac{48\times40+60\times40+48\times60}{40} = 180\Omega$$

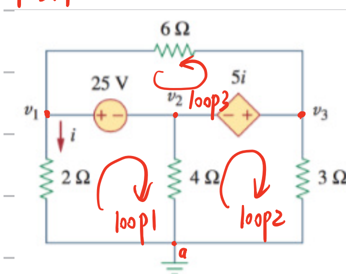
$$R_2 = \frac{48\times40+60\times40+48\times60}{60} = 120\Omega \qquad R_2' = \frac{120\times20}{120+20} = \frac{120}{7}\Omega$$

$$R_3 = \frac{48\times40+60\times40+48\times60}{48} = 150\Omega \qquad R_3' = \frac{150\times100}{150+100} = 60\Omega$$

$$R_{ab} = 6+\frac{180\times(\frac{120}{7}+60)}{180+\frac{120}{7}+60} = 6+54=60\Omega$$

$$i=\frac{240}{R_{ab}}=4A$$

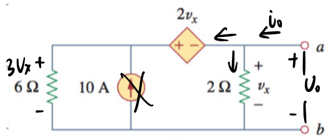
P 3.4



$$\begin{cases} 0-2i=V_1 \\ V_1-25=V_2 \\ V_2+5i=V_3 \\ i+\frac{V_2}{4}+\frac{V_3}{3}=0 \end{cases} \Rightarrow \begin{cases} i=3.804A \\ V_1=7.608V \\ V_2=-17.392V \\ V_3=1.628V \end{cases}$$

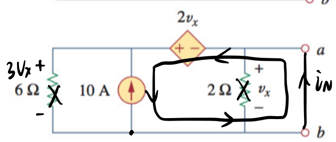
Week 2

4.12 诺顿定理



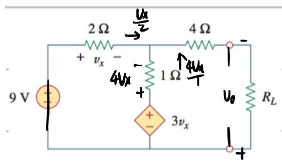
令 $U_{ab}=V_0$ ↻为正

$$\begin{cases} V_0=V_x \\ i_0=\frac{V_x}{2}+\frac{3V_x}{6} \end{cases} \qquad R_N=\frac{V_0}{i_0}=1\Omega$$

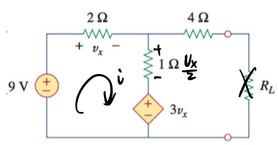


$i_N=10A$

4.13 最大功率传输



$$\begin{cases} V_0=V_x+4i_0=19V_x \\ i_0=\frac{V_x}{2}+\frac{4V_x}{1}=4.5V_x \end{cases} \qquad R_{TH}=\frac{V_0}{i_0}=4.222\Omega$$



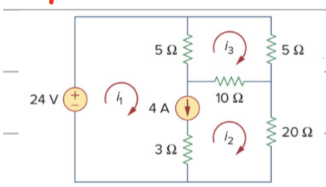
$-3V_x-\frac{V_x}{2}-V_x+9=0 \Rightarrow V_x=2V$

$V_{TH}=3V_x+\frac{V_x}{2}=7V$

☆2Ω和1Ω串 电压极性同

当 $R_L=R_{TH}$ 时 $P_{max}=\frac{V_{TH}^2}{4R_{TH}}=2.901W$

3.7

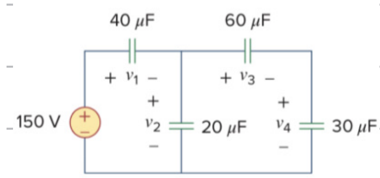


$$\begin{cases} -24+5(i_1-i_3)+10(i_2-i_3)+20i_2=0 \\ 5(i_3-i_1)+5i_3+10(i_3-i_2)=0 \\ i_1-i_2=4 \end{cases}$$

$$\Rightarrow \begin{bmatrix} 5 & 30 & -15 \\ -5 & -10 & 20 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 24 \\ 0 \\ 4 \end{bmatrix} \Rightarrow \begin{cases} i_1=4.8\,\boxed{A} \\ i_2=0.8\,\boxed{A} \\ i_3=1.6\,\boxed{A} \end{cases}$$

☆单位莫忘

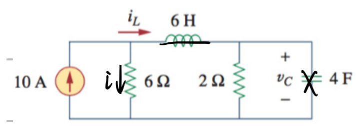
6.7 求 $v_1\ v_2\ v_3\ v_4$



$$\begin{cases} C_{34}=\frac{60\times30}{60+30}=20\mu F \\ C_{234}=20+20=40\mu F=C_1 \end{cases} \qquad \begin{cases} V_3+V_4=V_2 \\ 60V_3=30V_4 \end{cases}$$

∴ $V_2=V_1=\frac{1}{2}\times150=75V$ ∴ $V_3=25V$ $V_4=50V$

6.10 求 i_L V_c 和 L、C 储能 所设电路的 i、v 方向应与已知条件一致



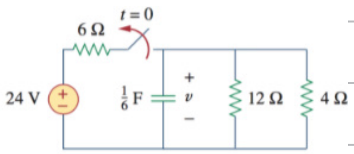
$$\begin{cases} 10=i+i_L \\ 6i=2i_L \end{cases} \Rightarrow i_L=7.5A \qquad V_c=2i_L=15V$$

$W_L=\frac{1}{2}Li_L^2=\frac{1}{2}\times6\times7.5^2=168.75J$

$W_c=\frac{1}{2}CV_c^2=\frac{1}{2}\times4\times15^2=450J$

Week 4

7.2



$R_{11} = \frac{12 \times 4}{12+4} = 3\,\Omega$

$V_0 = 24 \times \frac{3}{6+3} = 8\,V$

$R = R_{11} = 3\,\Omega$

$\tau = RC = \frac{1}{6} \times 3 = \frac{1}{2}$

$v(t) = V_0 e^{-\frac{t}{\tau}} = 8e^{-2t}\,(V)$

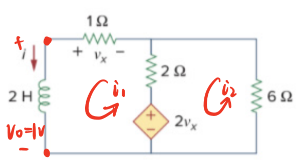
$p = vi = vC\frac{dv}{dt}$

$W_C(t) = \int_0^t p\,dt = \int_0^{v(t)} vC\,dv$

$= \frac{1}{2}Cv^2\Big|_0^{V_0}$

$= \frac{1}{2} \times \frac{1}{6} \times 8^2 = 5.333\,J$

7.3



施加激砺电压 $V_0 = 1V$

$$\begin{cases} 1 - 2V_x + 3i_1 - 2i_2 = 0 \\ 8i_2 - 2i_1 + 2V_x = 0 \\ -i_1 = V_x \end{cases} \Rightarrow \begin{cases} i_2 = -\frac{1}{8}A \\ i_1 = -\frac{1}{4}A \end{cases} \Rightarrow R = \left|\frac{V_0}{i_1}\right| = 4\,\Omega$$

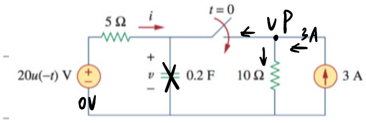
$\tau = L/R = 0.5$

$i(t) = i(0)e^{-\frac{t}{\tau}} = 7e^{-2t}\,(A)$

$V_x = -1 \times i(t) = -7e^{-2t}\,(V) \quad (t>0)$

Week 5

7.11 阶跃 RC



$$u(t)=\begin{cases}20 & t<0\\ 0 & t>0\end{cases}$$

$\forall t \quad u(t)=1 \quad u(-t)=1-u(t)$

$V_{(0)}=20\,U(t)=20V$

P节点电压 V : $\frac{V}{5}+\frac{V}{10}=3 \quad \Rightarrow V=10V$

$V_{(\infty)}=V-0=10V$

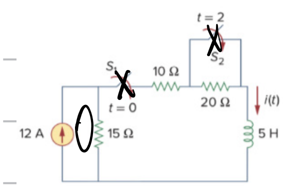
$\tau=RC=\frac{5\times10}{5+10}\times0.2=\frac{2}{3}$

$V_{(t)}=V_{(\infty)}+\left(V_{(0)}-V_{(\infty)}\right)e^{-\frac{t}{\tau}}=10\left(1+e^{-1.5t}\right)V \quad (t>0)$

$$\therefore V_{(t)}=\begin{cases}20V & t<0\\ 10\left(1+e^{-1.5t}\right)V & t>0\end{cases}$$

$$i_{(t)}=\begin{cases}0A & t<0\\ -2\left(1+e^{-1.5t}\right)V & t>0\end{cases}$$

7.13 阶跃 RL



$t=0$时关S_1，$t=2$时关S_2

算 $i_{(t)}$, $i_{(1)}$, $i_{(3)}$

分 $t\le0 \quad 0<t\le2 \quad t\ge2$ 3个阶段考虑

① 初态 $\quad i_{(0)}=i_{(0^-)}=i_{(0^+)}=0A$

② 关S_1后 零状态响应

$i_{(\infty)}=12\times\frac{15}{30+15}=4A \qquad \tau=\frac{L}{R}=\frac{5}{45}=\frac{1}{9}$

$i_{(t)}=i_{(\infty)}\left(1-e^{-\frac{t}{\tau}}\right)=4\left(1-e^{-9t}\right) \qquad (0<t<2)$

③ 关S_2后 $\quad i_{(2)}=4\left(1-e^{-18}\right)\approx i_{(0)}=4A$ 全响应

$i_{(\infty)}=12\times\frac{15}{15+10}=7.2A \qquad \tau=\frac{L}{R'}=\frac{5}{25}=\frac{1}{5}$

$i_{(t)}=i_{(\infty)}+\left(i_{(2)}-i_{(\infty)}\right)e^{-\frac{(t-2)}{\tau}}$

$=7.2+(4-7.2)e^{-5(t-2)}$

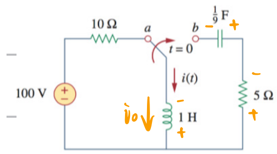
$=7.2-3.2e^{-5(t-2)} \qquad (t>2)$

$$\therefore i_{(t)}=\begin{cases}0 & t<0\\ 4\left(1-e^{-9t}\right) & 0<t<2\\ 7.2-3.2e^{-5(t-2)} & t>2\end{cases}$$

$i_{(1)}=4\left(1-e^{-9}\right)\approx4A$

$i_{(3)}\approx7.178A$

8.4 无源串联RLC



$i_{(0)}=\frac{100}{10}=10A \qquad V_{c(0)}=0V \neq L\left.\frac{di}{dt}\right|_{t=0}$

$V_{L(0)}+V_{C(0)}+R\,i_{(0)}=0 \quad \Rightarrow V_{L(0)}=-50V$

$\alpha=\frac{R}{2L}=\frac{5}{2\times1}=2.5 \qquad \omega_0=\frac{1}{\sqrt{LC}}=\frac{1}{\sqrt{1\times\frac{1}{9}}}=3$

$\alpha<\omega_0 \qquad \omega_d=\sqrt{\omega_0^2-\alpha^2}=1.658$

$i_{(t)}=e^{-2.5t}\left(B_1\cos1.658t+B_2\sin1.658t\right)$

$i_{(0)}=10 \Rightarrow B_1=10$

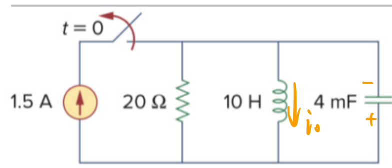
$\frac{di}{dt}=-2.5e^{-2.5t}\left(B_1\cos1.658t+B_2\sin1.658t\right)$

$\quad +1.658\,e^{-2.5t}\left(-B_1\sin1.658t+B_2\cos1.658t\right)$

$\Rightarrow -2.5B_1+1.658B_2=-50 \quad \Rightarrow B_2=15.076$

$\therefore i_{(t)}=e^{-2.5t}\left(10\cos1.658t-15.076\sin1.658t\right)$

8.6 无源并联RLC



V_0 i_0 按关联参考方向设

$i_{(0)}=1.5A=C\left.\frac{dv}{dt}\right|_{t=0} \qquad V_{(0)}=0V$

$\alpha=\frac{1}{2RC}=\frac{1}{2\times20\times4\times10^{-3}}=\frac{25}{4} \qquad \omega_0=\frac{1}{\sqrt{LC}}=5$

$\alpha>\omega_0 \qquad s_{1,2}=-\alpha\pm\sqrt{\alpha^2-\omega_0^2}=-2.5,\,-10$

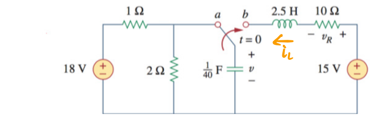
$V_{(t)}=A_1e^{-2.5t}+A_2e^{-10t}$

$$\begin{cases}V_{(0)}=A_1+A_2=0\\ \left.\frac{dV}{dt}\right|_{t=0}=\frac{-1.5}{4\times10^{-3}}=-2.5A_1-10A_2\end{cases} \Rightarrow \begin{cases}A_1=-50\\ A_2=50\end{cases}$$

$\therefore V_{(t)}=50\left(e^{-10t}-e^{-2.5t}\right)$

8.7 阶跃串联RLC

Having been in position a for a long time, the switch in Fig. 8.21 is moved to position b at $t = 0$. Find $v(t)$ and $v_R(t)$ for $t > 0$.



$i_L(0) = 0A \quad V_c(0) = 18 \times \frac{2}{1+2} = 12V$

$\alpha = \frac{R}{2L} = \frac{10}{2\times2.5} = 2 \quad W_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2.5\times\frac{1}{40}}} = 4$

$\alpha < W_0 \quad W_d = \sqrt{W_0^2 - \alpha^2} = 3.464$

$V(t) = V_s + e^{-2t}(B_1\cos 3.464t + B_2 \sin 3.464t) \quad V_s = 15V$

$V(0) = 12 = 15 + B_1 \Rightarrow B_1 = -3$

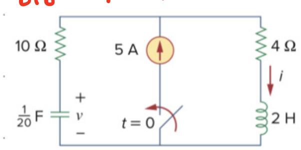
$i(0) = C\frac{dV}{dt}\Big|_{t=0} = -2B_1 + 3.464B_2 = 0 \Rightarrow B_2 = -1.732$

$\therefore V(t) = 15 - e^{-2t}(3\cos 3.464t + 1.7321 \sin 3.464t)$

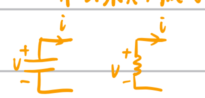
$i(t) = C\frac{dV}{dt} = 0.3464 \sin 3.464t \, e^{-2t} A$

$V_R(t) = -Ri(t) = 3.464 e^{-2t} \sin 3.464t \ V$

8.8 阶跃并联RLC



算出来负则反向



$$\begin{cases} 10\times C\frac{dv}{dt} + v = 4i + L\frac{di}{dt} \\ C\frac{dv}{dt} + i = 5 \end{cases} \Rightarrow \frac{d^2i}{dt^2} + 7\frac{di}{dt} + 10i = 50$$

$\Rightarrow 7 = \frac{1}{RC} \quad 10 = \frac{1}{LC} \quad 50 = \frac{I_s}{LC}$

$i_L(0) = 0A \quad V_c(0) = 0V \qquad V_c(0) + 5\times10 = 4\times0 + V_L(0) \Rightarrow V_L(0) = 50V$

$\alpha = \frac{1}{2RC} = 3.5 \quad W_0 = \frac{1}{\sqrt{LC}} = 3.1623$

$\alpha > W_0 \quad S_{1,2} = -\alpha \pm \sqrt{\alpha^2 - W_0^2} = -2, -5$

$i(t) = I_s + A_1 e^{-2t} + A_2 e^{-5t} \qquad I_s = 5A$

$i(0) = 5 + A_1 + A_2 = 0$

$L\frac{di}{dt}\Big|_{t=0} = \frac{V_L(0)}{2} = -2A_1 - 5A_2 = 25$

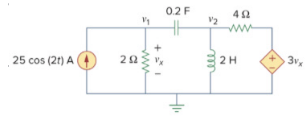
$\Rightarrow \begin{cases} A_1 = 0 \\ A_2 = -5 \end{cases}$

$\therefore i(t) = 5(1 - e^{-5t}) \ A$

$V(t) = \int_0^t C(5 - i(t))\,dt = 20(1 - e^{-5t}) V$

10.1 频域节点电压

$w = 2$



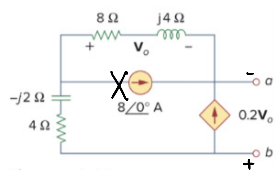
$Z_L = jWL = j4 \qquad Z_C = \frac{1}{jWC} = -j2.5$

$25\cos 2t \Rightarrow 25\angle 0°$

$$\begin{cases} V_1点: \ 25\angle0° + \frac{0-V_1}{2} + \frac{V_2 - V_1}{-2.5j} = 0 \\ V_2点: \ \frac{V_1 - V_2}{-2.5j} + \frac{3V_x - V_2}{4} + \frac{0 - V_2}{4j} = 0 \\ V_1 = V_x \end{cases} \Rightarrow \begin{cases} -(0.5+0.4j)V_1 + 0.4jV_2 = -25 \\ (0.75+0.4j)V_1 - (0.75+0.15j)V_2 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} V_1 = 28.31\angle 60.01° \\ V_2 = 82.56\angle 57.12° \end{cases} \Rightarrow \begin{cases} V_1(t) = 28.31\cos(2t + 60.01°) V \\ V_2(t) = 82.56\cos(2t + 57.12°) V \end{cases}$$

10.9 频域戴维南定理



$$\begin{cases} V_{ba} = V_0 + \frac{4-j2}{8+j4}V_0 = (1.3 - 0.4j)V_0 \\ I_{ba} = -0.2V_0 - \frac{V_0}{8+j4} = (0.3 - 0.05j)V_0 \end{cases} \Rightarrow Z_{Th} = \frac{V_{ba}}{I_{ba}} = 4.473\angle{-7.64°} \ \Omega$$

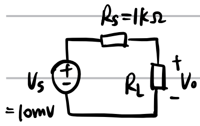
$V_{Th} = V_0 + \frac{4-j2}{8+j4}V_0 = 11.763\angle 12.9° \ V$

Week 6/7

Lecture 8

1.3 A signal source that is most conveniently represented by its Thévenin equivalent has $v_s = 10$ mV and $R_s = 1$ kΩ. If the source feeds a load resistance R_L, find the voltage v_o that appears across the load for $R_L = 100$ kΩ, 10 kΩ, 1 kΩ, and 100 Ω. Also, find the lowest permissible value of R_L for which the output voltage is at least 80% of the source voltage.
Ans. 9.9 mV; 9.1 mV; 5 mV; 0.9 mV; 4 kΩ

$$V_o = \frac{R_L}{R_s+R_L}V_s$$



$R_L = 100k\Omega$ $V_o = 9.9mV$

$R_L = 10k\Omega$ $V_o = 9.1mV$

$R_L = 1k\Omega$ $V_o = 5mV$

$R_L = 100\Omega$ $V_o = 0.9mV$

$V_o \geq 80\% V_s \Rightarrow R_L \geq 4k\Omega$

D1.24 Consider the situation illustrated in Fig. 1.27. Let the output resistance of the first voltage amplifier be 1 kΩ and the input resistance of the second voltage amplifier (including the resistor shown) be 9 kΩ. The resulting equivalent circuit is shown in Fig. E1.24 where V_s and R_s are the output voltage and output resistance of the first amplifier, C is a coupling capacitor, and R_i is the input resistance of the second amplifier. Convince yourself that V_2/V_s is a high-pass STC function. What is the smallest value for C that will ensure that the 3-dB frequency is not higher than 100 Hz?
Ans. 0.16 μF

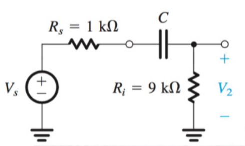


Figure E1.24

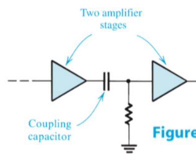


Figure 1.27

$$V_2 = V_s\frac{R_i}{R_s+R_i+\frac{1}{j\omega C}}$$

$$\Rightarrow \frac{V_2}{V_s} = \frac{R_i/(R_s+R_i)}{1-j\frac{1}{\omega C(R_s+R_i)}} = \frac{K}{1-j(\omega_0/\omega)}$$

$$\therefore \omega_0 = \frac{1}{C(R_s+R_i)} \leq 2\pi\times100 \qquad \therefore C \geq 0.16\mu F$$

注意单位换算
ω 单位 rad/s
f 单位 Hz
$\omega = 2\pi f$

Lecture 9

2.6 For the circuit in Fig. E2.6 determine the values of v_1, i_1, i_2, v_O, i_L, and i_O. Also determine the voltage gain v_O/v_I, current gain i_L/i_I, and power gain P_O/P_I.
Ans. 0 V; 2 mA; 2 mA; −10 V; −10 mA; −12 mA; −5 V/V (14 dB); −5 A/A (14 dB); 25 W/W (14 dB)

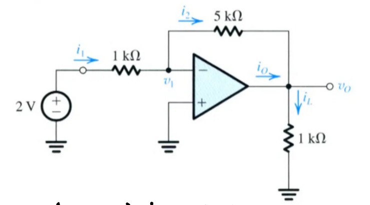


Figure E2.6

$i_1 = \frac{2}{1000} = 2mA$ $\quad i_2 = i_1 + 0 = 2mA$

$V_1 = 0V$ $\quad V_o = 0 - 5\times2 = -10V$ $\quad i_L = \frac{-10}{1\times10^3} = -10mA$ $\quad i_o = i_L - i_2 = -10-2 = -12mA$

? $\frac{V_o}{V_i} = \frac{-10}{2} = -5\,V/V$ $\quad \frac{i_o}{i_1} = \frac{-10}{2} = -5\,A/A$ $\quad \frac{P_o}{P_1} = \frac{V_o i_o}{V_i i_1} = \frac{10\times10}{2\times2} = 25\,W/W$

D2.8 Use the idea presented in Fig. 2.11 to design a weighted summer that provides

$$v_O = 2v_1 + v_2 - 4v_3$$

10 kW 是常用电阻

Ans. A possible choice: $R_1 = 5$ kΩ, $R_2 = 10$ kΩ, $R_a = 10$ kΩ, $R_b = 10$ kΩ, $R_3 = 2.5$ kΩ, $R_c = 10$ kΩ

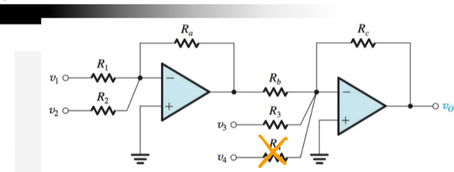


Figure 2.11 A weighted summer capable of implementing summing coefficients of both signs.

$\frac{R_a}{R_1}\cdot\frac{R_c}{R_b} = 2$ $\quad \frac{R_c}{R_3} = 4$ $\quad$ 令$R_a = R_b = R_c = 10k\Omega$

$\frac{R_a}{R_2}\cdot\frac{R_c}{R_b} = 1$ $\quad$ 则 $R_3 = 2.5k\Omega$

$R_2 = 10k\Omega$ $\quad R_1 = 5k\Omega$

D2.16 Find values for the resistances in the circuit of Fig. 2.16 so that the circuit behaves as a difference amplifier with an input resistance of 20 kΩ and a gain of 10.
Ans. $R_1 = R_3 = 10$ kΩ; $R_2 = R_4 = 100$ kΩ

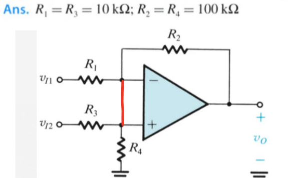


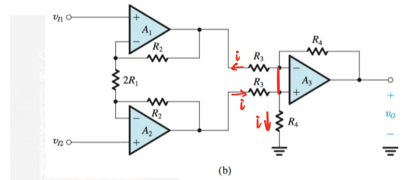
Figure 2.16 A difference amplifier.

$R_i = R_1 + R_3 = 20k\Omega$ $\quad$ 令$R_1 = R_3$ $\quad R_2 = R_4$

$$V_o = V_{i1}\left(-\frac{R_2}{R_1}\right) + \left(\frac{R_4}{R_3+R_4}\right)\left(1+\frac{R_2}{R_1}\right)V_{i2} \Rightarrow \frac{V_o}{V_{id}} = \frac{R_2}{R_1} = 10$$

$\therefore R_1 = R_3 = 10k\Omega$ $\quad R_2 = R_4 = 100k\Omega$

2.17 Consider the instrumentation amplifier of Fig. 2.20(b) with a common-mode input voltage of +5 V (dc) and a differential input signal of 10-mV-peak sine wave. Let $(2R_1) = 1$ kΩ, $R_2 = 0.5$ MΩ, and $R_3 = R_4 = 10$ kΩ. Find the voltage at every node in the circuit.
Ans. $v_{I1} = 5 - 0.005\sin\omega t$; $v_{I2} = 5 + 0.005\sin\omega t$; v_- (op amp A_1) $= 5 - 0.005\sin\omega t$; v_- (op amp A_2) $= 5 + 0.005\sin\omega t$; $v_{O1} = 5 - 5.005\sin\omega t$; $v_{O2} = 5 + 5.005\sin\omega t$; $v_-(A_3) = v_+(A_3) = 2.5 + 2.5025\sin\omega t$; $v_O = 10.01\sin\omega t$ (all in volts)



$V_{i1} = 5 - 0.005\sin\omega t = V_-(\text{op amp } A_1)$ $\quad V_{o1} = V_{i1} - \frac{V_{id}}{2R_1}R_2 = 5 - 5.005\sin\omega t$

$V_{i2} = 5 + 0.005\sin\omega t = V_-(\text{op amp } A_2)$ $\quad V_{o2} = V_{i2} + \frac{V_{id}}{2R_1}R_2 = 5 + 5.005\sin\omega t$

$V_-(A_3) = \frac{V_{o2}+0}{2} = 2.5 + 2.5025\sin\omega t = V_+(A_3)$

$V_o = \frac{R_4}{R_3}\left(1+\frac{R_2}{R_1}\right)V_{id} = 10.01\sin\omega t$

Week 8

3.3 For a silicon crystal doped with 硼 $\boxed{\text{boron}}$ what must N_A be if at $T = 300$ K the electron concentration drops below the intrinsic level by a factor of 10^6?
Ans. $N_A = 1.5 \times 10^{16}/\text{cm}^3$

$\boxed{p_p \simeq N_A \quad p_p n_p = n_i^2 \quad n_i \simeq 1.5\times10^{10}/\text{cm}^3}$

$$n_i = BT^{3/2} e^{-E_g/2kT} = 1.5\times10^{10}\ \text{carries/cm}^3 \quad (\text{for } T=300k)$$

$$P_p \approx N_A \quad P_p n_p = n_i^2 \quad n_p = 10^{-6} n_i \Rightarrow N_A = 1.5\times10^{16}/\text{cm}^3$$

A diode operates in the forward bias region with a typical current level [i.e., $I_D \approx I_S \exp(V_D/V_T)$]. Suppose we wish to increase the current by a factor of 10. How much change in V_D is required?

Ans. V_D changes by 60 mV for a decade (tenfold) change in I_D

$$\frac{I_D'}{I_D} = \frac{I_s e^{V_D'/V_T}}{I_s e^{V_D/V_T}} = e^{\frac{V_D'-V_D}{V_T}} = 10 \qquad V_D'-V_D = 26\ln 10 = 60\text{mV}$$

知识点

1. 载流子 $P_p = n_i + N_A \approx N_A \qquad P_p n_p = n_i^2$ 质量作用定律

$n_n = n_i + N_D \approx N_D \qquad P_n n_n = n_i^2$

2. 漂移电流 $\begin{cases} \vec{V_h} = \mu_p \vec{E} \\ \vec{V_e} = -\mu_n \vec{E} \end{cases}$ 迁移率 $\xrightarrow{\text{饱和}} \begin{cases} \mu = \frac{\mu_0}{1+bE} \\ V_{sat} = \frac{\mu_0}{b} \\ V = \frac{\mu_0}{1+\frac{\mu_0 E}{V_{sat}}} E \end{cases}$

$$\begin{cases} I_p = PqAV_h & J_p = qP\mu_p E \\ I_n = nqAV_e & J_n = qn\mu_n E \end{cases}$$

$$\begin{cases} J = J_p + J_n = q(P\mu_p + n\mu_n)E \\ J = \sigma E \end{cases} \Rightarrow \begin{cases} \sigma = q(p\mu_p + n\mu_n) \\ \rho = q(p\mu_p + n\mu_n) \end{cases}$$

3. 扩散电流 $\begin{cases} J_p = -qD_p \frac{dP(x)}{dx} \\ J_n = qD_n \frac{dn(x)}{dx} \end{cases} \Rightarrow \frac{D}{\mu} = \frac{KT}{q} = V_T \approx 26\text{mV} @300k$

4. 内建电势 $\begin{cases} E = -\frac{dV}{dx} \\ qP\mu_p E = qD_p \frac{dP}{dx} \end{cases} \Rightarrow -\mu_p \int_{x_1}^{x_2} dV = D_p \int_{P_n}^{P_p} \frac{dP}{P}$

$\Uparrow$

$I_D = I_s$

$\Rightarrow \left.\begin{aligned} V_{x_2} - V_{x_1} = -\frac{D_p}{\mu_p} \ln \frac{P_p}{P_n} \\ \frac{D}{\mu} = \frac{kT}{q} \end{aligned}\right\} |V_0| = \frac{kT}{q} \ln \frac{P_p}{P_n} = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2}$

5. 耗尽区宽度 $W = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right) V_{bi}}$

ε_s 半导体介电常数 $\quad V_{bi}$ 内建电势

6.

	正偏	反偏
外电源+极	p	n
耗尽区	窄	宽
扩散电流	↑	↓
漂移电流	很小	不变
电容效应	扩散电容 $C_d = \frac{dQ}{dV}$ (移动载流子)	结电容 $C_j = \frac{C_{j0}}{\sqrt{1+\frac{V_R}{V_0}}}$ $C_{j0} = \sqrt{\frac{\varepsilon_{si}}{2} \frac{N_A N_D}{N_A + N_D} \frac{1}{V_0}}$ (耗尽区固定离子)

7. $\begin{cases} \text{齐纳击穿：高掺杂 — 耗尽区E大 — 共价键 — 电子空穴对} \quad <5V \\ \text{雪崩击穿：低掺杂 — 反向电压大 — 耗尽区中性原子 — 电离} \quad >7V \end{cases}$

8. PN结 $I = I_s(e^{V/V_T} - 1)$

9. 热电压 V_T (热运动给载流子带来的等效电压)

热平衡下，粒子能量玻尔兹曼分布 $n(E) \propto e^{-E/kT} \quad E = qV$

$\Rightarrow$ 定义 $V_T = \frac{kT}{q} \quad \frac{n_1}{n_2} = e^{-\frac{(V_1 - V_2)}{V_T}}$

Week 9

作业

*3.9. Plot the input/output characteristics of the circuits depicted in Fig. 3.69 using an ideal model for the diodes. Assume $V_B = 2$ V.

*3.10. Repeat Problem 3.9 with a constant-voltage diode model. $V_{D,on} = 0.8V$

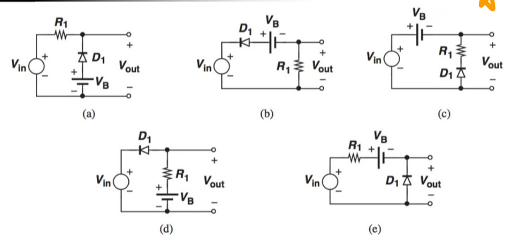
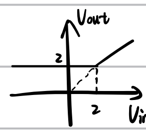


Figure 3.69

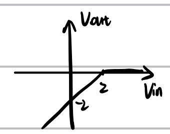
理想模型

Vin 从 $-\infty \to +\infty$

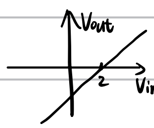
判二极管状态转折点



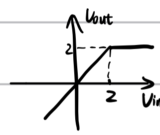
(a)



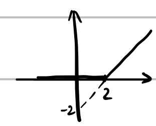
(b)



(c)



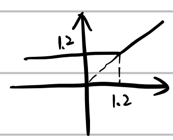
(d)



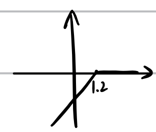
(e)

D通断不影响 Vout

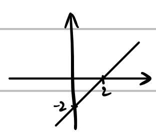
恒压降模型



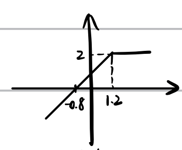
(a)



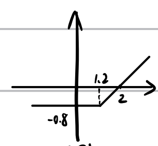
(b)



(c)



(d)

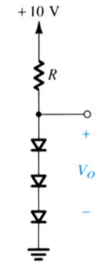


(e)

D4.13 Design the circuit in Fig. E4.13 to provide an output voltage of 2.4 V. Assume that the diodes available have 0.7-V drop at 1 mA.

Ans. $R = 139\ \Omega$

V_T 取 25mV



$$\begin{cases} I_D = I_S \cdot e^{V_D/V_T} = I_0 e^{(V_D - V_0)/V_T} = 0.0546\ A \\ I_D = \dfrac{V_{DD} - 3V_D}{R} \Rightarrow R = 139\,\Omega \end{cases}$$

$3V_D = V_o = 2.4 \Rightarrow V_D = 0.8$

$I_0 = I_S \cdot e^{V_0/V_T} = 1mA$

Figure E4.13

Week 10

5.1 A 0.18-μm fabrication process is specified to have $t_{ox} = 4$ nm, $\mu_n = 450\ \text{cm}^2/\text{V}\cdot\text{s}$, and $V_t = 0.5$ V. Find the value of the process transconductance parameter k_n'. For a MOSFET with minimum length fabricated in this process, find the required value of W so that the device exhibits a channel resistance r_{DS} of 1 kΩ at $v_{GS} = 1$ V.
Ans. 388 μA/V²; 0.93 μm

$\varepsilon_{ox} = 3.45 \times 10^{-11}\ F/m$

$$k_n' = \mu_n C_{ox} = \mu_n \frac{\varepsilon_{ox}}{t_{ox}} = 450 \times 10^{-4} \times \frac{8.85 \times 10^{-12}}{4 \times 10^{-9}} = 388.125\ \mu A/V^2$$

$$r_{DS} = \frac{V_{DS}}{i_D} = \frac{1}{(\mu_n C_{ox})\left(\frac{W}{L}\right) V_{ov}}$$

$$\Rightarrow W = \frac{L}{k_n' r_{DS} V_{ov}} = \frac{0.18 \times 10^{-6}}{388.125 \times 10^{-6} \times 1 \times 10^3 \times (1-0.5)} = 0.928\ \mu m$$

5.6 An NMOS transistor is fabricated in a 0.18-μm process having $\mu_n C_{ox} = 400\ \mu\text{A/V}^2$ and $V_A' = 5$ V/μm of channel length. If $L = 0.8$ μm and $W = 16$ μm, find V_A and λ. Find the value of i_D that results when the device is operated with an overdrive voltage $v_{OV} = 0.2$ V and $v_{DS} = 0.8$ V. Also, find the value of r_o at this operating point. If v_{DS} is increased by 1 V, what is the corresponding change in i_D?
Ans. 4 V; 0.25 V⁻¹; 192 μA; 25 kΩ; 40 μA

$$V_A = V_A' L = 0.8 \times 5 = 4V \qquad \lambda = \frac{1}{V_A} = 0.25\ V^{-1}$$

$$i_D = \frac{1}{2} k_n' \left(\frac{W}{L}\right) (V_{GS} - V_{th})^2 (1 + \lambda V_{DS}) = \frac{1}{2} \times 400 \times \frac{16}{0.8} \times (0.2)^2 \times \left(1 + \frac{1}{4} \times 0.8\right) = 192\ \mu A$$

$$I_D = \frac{1}{2} k_n' \left(\frac{W}{L}\right) (V_{GS} - V_{th})^2 = 160\ \mu A \qquad r_o = \frac{V_A}{I_D} = 25\ k\Omega$$

$$i_D' = \frac{1}{2} \times 400 \times \frac{16}{0.8} \times (0.2)^2 \times \left(1 + \frac{1}{4} \times 1.8\right) = 232\ \mu A \qquad \Delta i_D = i_D' - i_D = 40\ \mu A$$

D5.8 Redesign the circuit of Fig. 5.21 for the following case: $V_{DD} = -V_{SS} = 1$ V, $V_t = 0.4$ V, $\mu_n C_{ox} = 400\ \mu\text{A/V}^2$, $W/L = 5\ \mu\text{m}/0.4\ \mu\text{m}$, $I_D = 100\ \mu$A, and $V_D = +0.2$ V.
Ans. $R_D = 8$ kΩ; $R_S = 4$ kΩ

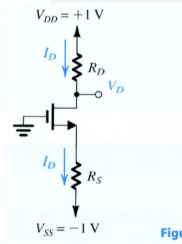


Figure 5.21 Circuit for Example 5.3.

$$R_D = \frac{V_{DD} - V_D}{I_D} = 8\ k\Omega$$

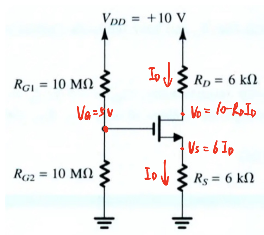
$\because V_D > V_G - V_{Th} \Rightarrow V_{DS} > V_{GS} - V_{Th}$ 工作在饱和区

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{ov}^2 = 100\ \mu A$$

$\Rightarrow V_{ov} = V_{GS} - V_t = 0.2V \quad \therefore V_G - V_S = -V_S = 0.6V$

$$R_S = \frac{V_S - V_{SS}}{I_D} = \frac{-0.6+1}{100\mu} = 4k\Omega$$

5.12 For the circuit of Fig. 5.24, what is the largest value that R_D can have while the transistor remains in the saturation mode? Let $V_{tn} = 1$ V and $k_n'(W/L) = 1\ \text{mA/V}^2$
Ans. 12 kΩ



$V_{GS} > V_{Th}$

$V_{DS} > V_{GS} - V_{tn} \Leftrightarrow 10 - R_D I_D > 5 - 1$

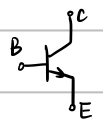
$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{tn})^2$$

$$= \frac{1}{2} \times 1 \times (5 - 6I_D - 1)^2 \quad \Rightarrow I_D \approx 0.5mA$$

$$\therefore R_D < \frac{6}{I_D} = 12k\Omega$$

Week 11

6.5 A transistor for which $I_S = 10^{-16}$ A and $\beta = 100$ is conducting a collector current of 1 mA. Find v_{BE}. Also, find I_{SE} and I_{SB} for this transistor.
Ans. 747.5 mV; 1.01×10^{-16} A; 10^{-18} A



假设在放大区　取 $V_T = 25mV$

$I_C = I_S e^{V_{BE}/V_T} \Rightarrow V_{BE} = V_T \ln \frac{I_C}{I_S} = 748.34mV$

$I_{SE} = \frac{\beta+1}{\beta} I_S = 1.01 \times 10^{-16}A$　　$I_E = I_{SE} e^{V_{BE}/V_T}$

$I_{SB} = \frac{1}{\beta} I_S = 10^{-18}A$　　$I_B = I_{SB} e^{V_{BE}/V_T}$

6.13 In the circuit shown in Fig. E6.13, the voltage at the emitter was measured and found to be –0.7 V. If $\beta = 50$, find I_E, I_B, I_C, and V_C.

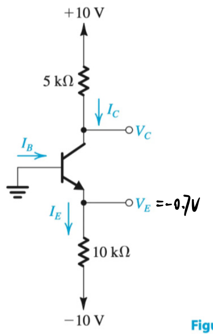


Figure E6.13

$I_E = \frac{V_E - (-10)}{10k} = 0.93mA$

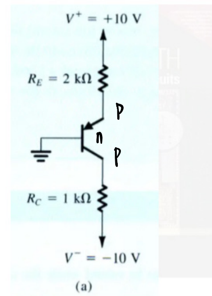
$I_B = I_E \times \frac{1}{51} = 18.2\mu A$

$I_C = I_E \times \frac{50}{51} = 0.91\ mA$

$V_C = 10 - I_C \cdot 5k = 5.45V$

Ans. 0.93 mA; 18.2 μA; 0.91 mA; +5.45 V

D6.25 For the circuit in Fig. 6.26(a), find the largest value to which R_C can be raised while the transistor remains in the active mode.
Ans. 2.24 kΩ



$V_{EB} = 0.7V$　$V_E = 0.7V$

$I_E = \frac{V^+ - V_E}{R_E} = \frac{10-0.7}{2k} = 4.65mA$

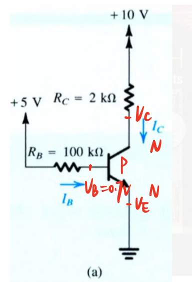
设$\beta = 100$　$I_C = \alpha I_E = 4.60mA$　直接取 $I_C = I_E$

$V_C - V^- = I_C R_C$

$V_C = V^- + I_C R_C < V_B + 0.4$

$\Rightarrow R_C \le 2.26k\Omega$　$2.24k\Omega$

D6.27 The circuit of Fig. 6.27(a) is to be fabricated using a transistor type whose β is specified to be in the range of 50 to 150. That is, individual units of this same transistor type can have β values anywhere in this range. Redesign the circuit by selecting a new value for R_C so that all fabricated circuits are guaranteed to be in the active mode. What is the range of collector voltages that the fabricated circuits may exhibit?
Ans. $R_C = 1.5$ kΩ; $V_C = 0.3$ V to 6.8 V



假设工作在放大区

$I_B = \frac{5-0.7}{100k} = 0.043mA$

$V_C = 10 - I_C R_C > V_B - 0.4$

当$\beta = 150$时　$I_C = \beta I_B = 6.45mA$

$V_B = 5 - 100k \cdot I_B = 0.7V$　$R_C = 1.50k\Omega$

$V_C = 0.3V$

当$\beta = 50$时　$I_C = 2.15mA$　$V_C = 10 - I_C R_C = 6.78V$

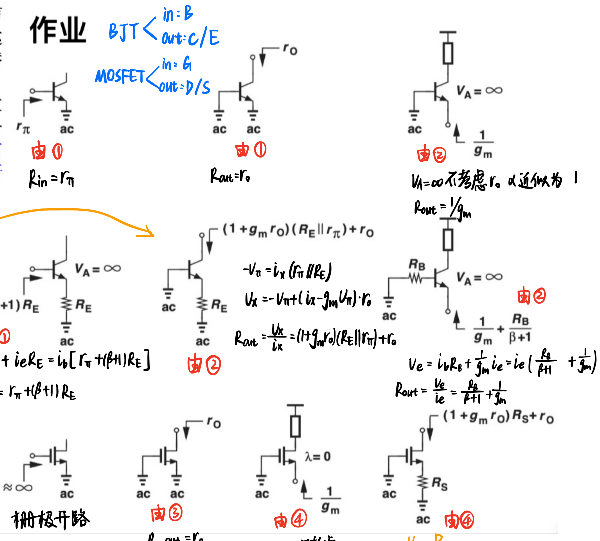
Input and Output Impedances

验证常见结构的小信号输入输出电阻表达式（用小信号模型替代MOSFET/BJT，再求输入输出电阻，求输出电阻时注意信号源短路。熟记该图有利于快速直观地分析MOSFET/BJT电路）

作业

BJT < in: B; out: C/E

MOSFET < in: G; out: D/S



由①　$R_{in} = r_\pi$

由①　$R_{out} = r_o$

由②　$V_A = \infty$ 不考虑 r_o　α 近似为 1

$R_{out} = 1/g_m$

由①　$V_b = i_b r_\pi + i_e R_E = i_b[r_\pi + (\beta+1)R_E]$

$R_{in} = \frac{V_b}{i_b} = r_\pi + (\beta+1)R_E$

由②　$-V_\pi = i_x (r_\pi // R_E)$

$U_x = -V_\pi + (i_x - g_m V_\pi) \cdot r_o$

$R_{out} = \frac{V_x}{i_x} = (1+g_m r_o)(R_E \| r_\pi) + r_o$

由②　$V_e = i_b R_B + \frac{1}{g_m} i_e = i_e\left(\frac{R_B}{\beta+1} + \frac{1}{g_m}\right)$

$R_{out} = \frac{V_e}{i_e} = \frac{R_B}{\beta+1} + \frac{1}{g_m}$

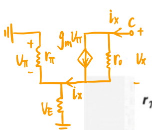
栅极开路

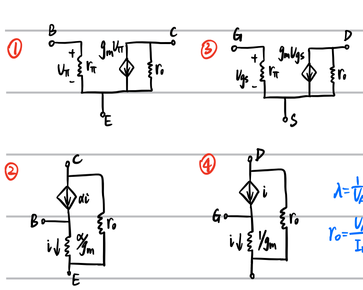
由③　$R_{out} = r_o$

由④　$\lambda = 0$ 不考虑 r_o

$R_{out} = \frac{1}{g_m}$

由④

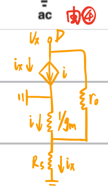




$r_o = \frac{V_A}{I_C'}$

$\lambda = \frac{1}{V_A}$

$r_o = \frac{V_A}{I_D'}$

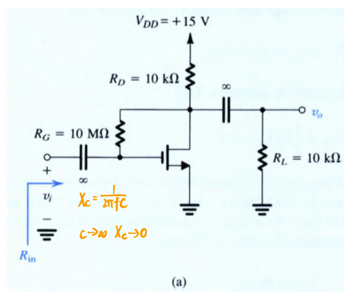


$V_x = i_x R_s + (i_x - i) r_o$

$-i = \frac{i_x R_s}{1/g_m}$

$\Rightarrow R_{out} = \frac{V_x}{i_x} = R_s + r_o + g_m R_s r_o$

D7.10 Consider the amplifier circuit of Fig. 7.16(a) without the load resistance R_L and with channel-length modulation neglected. Let $V_{DD} = 5$ V, $V_t = 0.7$ V, and $k_n = 1$ mA/V². Find V_{OV}, I_D, R_D, and R_G to obtain a voltage gain of –25 V/V and an input resistance of 0.5 M Ω. What is the maximum allowable input signal, $\hat{v}_i$?
Ans. 0.319 V; 50.9 μA; 78.5 kΩ; 13 MΩ; 27 mV



$R_{in} = \frac{R_G}{1 + g_m R_D} = 0.5M\Omega$
$A_v = g_m R_D = 25$
$\Big\}$ $R_G = 13M\Omega$

$I_D = \frac{1}{2} k_n V_{OV}^2$
$k_n V_{OV} = \frac{A_v}{R_D} = \frac{25}{R_D}$
$V_{OV} = 5 - I_D R_D - 0.7$
$\Big\} \Rightarrow V_{OV} = 0.319V$
$\therefore I_D = 50.9\mu A$
$R_D = 78.5k\Omega$

$V_{i,max} = \frac{V_t}{|A_v|+1} = \frac{0.7}{26} = 27mV$

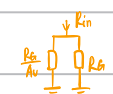
$A_v = -g_m R_D$　　$A_v = -g_m R_C$

$i_D = I_D + \mu_n C_{ox} \frac{W}{L} V_{OV} V_{gs}$　　$i_C = I_C + \frac{I_C}{V_T} V_{be}$

$g_m = \mu_n C_{ox} \frac{W}{L} V_{OV} = k_n V_{OV}$　　$g_m = \frac{I_C}{V_T}$

$g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}$

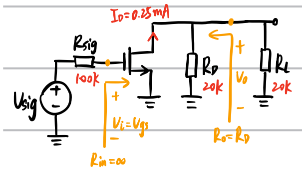
$g_m = \frac{I_D}{V_{OV}/2}$



$\frac{R_G}{\frac{R_G}{A_v}} \cdot \frac{R_G}{A_v}$... $\frac{R_G^2}{A_v} \big/ \left(\frac{1}{A_v}+1\right)R_G = \frac{R_G}{1+A_v}$

Week 12

7.21 A CS amplifier utilizes a MOSFET biased at $I_D = 0.25$ mA with $V_{OV} = 0.25$ V and $R_D = 20$ kΩ. The amplifier is fed with a signal source having $R_{sig} = 100$ kΩ, and a 20-kΩ load is connected to the output. Find R_{in}, A_{vo}, R_o, A_v, and G_v. If, to maintain reasonable linearity, the peak of the input sine-wave signal is limited to 10% of $2V_{OV}$, what is the peak of the sine-wave voltage at the output?
Ans. ∞; −40 V/V; 20 kΩ; −20 V/V; −20 V/V; 1 V



$g_m = \frac{I_D}{V_{ov}/2} = \frac{0.25mA}{0.25/2\,V} = 2m\Omega$

$R_{in} = \infty \qquad R_o = R_D = 20k\Omega$

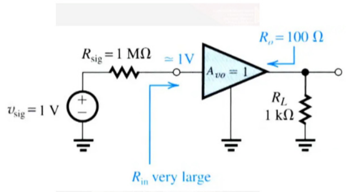
$A_{vo} = -g_m R_D = -40\,V/V$

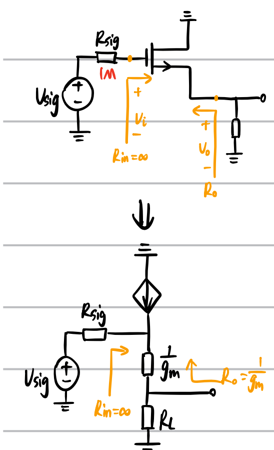
$A_v = A_{vo}\frac{R_L}{R_L + R_o} = -g_m(R_L \| R_D) = -20\,V/V$

$G_v = \frac{V_o}{V_{sig}} = -g_m(R_D \| R_L) = -20\,V/V$

$V_{sig} = 10\% \times 2 \times 0.25 = 0.05V \qquad V_o = |G_v|\, V_{sig} = 1V$

D7.28 It is required to design a source follower that implements the buffer amplifier shown in Fig. 7.42(c). If the MOSFET is operated with an overdrive voltage $V_{OV} = 0.25$ V, at what drain current should it be biased? Find the output signal amplitude and the signal amplitude between gate and source.
Ans. 1.25 mA; 0.91 V; 91 mV





$g_m = \frac{I_D}{V_{ov}/2}$

$R_{in} = \infty \qquad R_o = \frac{1}{g_m} = 100\Omega \Rightarrow I_D = 1.25mA$

$A_{vo} = 1$

$A_v = \frac{V_o}{V_i} = \frac{R_L}{R_L + 1/g_m} = \frac{1000}{1000+100} = 0.91$

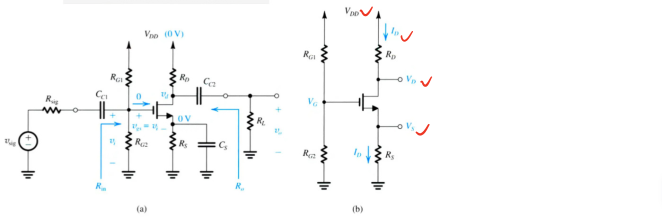
$G_v = A_v$

$V_o = V_{in}|G_v| = 0.91V$

$V_{gs} = \frac{\frac{1}{g_m}}{R_L} V_o = 91mV$

Week 13

D7.38 Design the bias circuit in Fig.7.57(b) for the CS amplifier of Fig. 7.57(a). Assume the MOSFET is specified to have $V_t = 1$ V, $k_n = 4$ mA/V^2, and $V_A = 100$ V. Neglecting the Early effect, design for $I_D = 0.5$ mA, $V_S = 3.5$ V, and $V_D = 6$ V using a power-supply $V_{DD} = 15$ V. Specify the values of R_S and R_D. If a current of 2 μA is used in the voltage divider, specify the values of R_{G1} and R_{G2}. Give the values of the MOSFET parameters g_m and r_o at the bias point.
Ans. $R_S = 7$ kΩ; $R_D = 18$ kΩ; $R_{G1} = 5$ MΩ; $R_{G2} = 2.5$ MΩ; $g_m = 2$ mA/V; $r_o = 200$ kΩ



$$R_D = \frac{V_{DD}-V_D}{I_D} = \frac{(15-6)V}{0.5mA} = 18k\Omega$$

$$R_S = \frac{V_S}{I_D} = \frac{3.5V}{0.5mA} = 7k\Omega$$

$$I_D = \frac{1}{2}k_n(V_{GS}-V_t)^2 \Rightarrow \frac{1}{2} = \frac{1}{2}\times 4\times(V_{GS}-1)^2 \Rightarrow V_{GS} = 1.5V$$

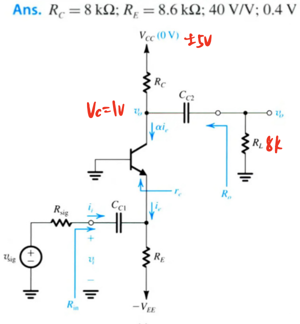
$$V_G = V_{GS}+V_S = 1.5+3.5 = 5V$$

$$R_{G1}+R_{G2} = \frac{V_{DD}}{I_d} = \frac{15}{2\times10^{-6}} = 7500k\Omega$$

$$V_G = \frac{R_{G2}}{R_{G1}+R_{G2}}V_{DD} \Rightarrow R_{G1} = 5M\Omega,\ R_{G2} = 2.5M\Omega$$

$$g_m = \frac{I_D}{V_{ov}/2} = \frac{0.5mA}{(1.5-1)/2} = 2mA/V \qquad r_o = \frac{V_A}{I_D} = \frac{100V}{0.5mA} = 200k\Omega$$

D7.44 Design the CB amplifier of Fig. 7.60(a) to provide an input resistance R_{in} that matches the source resistance of a cable (电缆) with a characteristic resistance of 50 Ω. Assume that $R_E \gg r_e$. The available power supplies are ±5 V and $R_L = 8$ kΩ. Design for a dc collector voltage $V_C = +1$ V. Specify the values of R_C and R_E. What is the overall voltage gain? If v_{sig} is a sine wave with a peak amplitude of 10 mV, what is the peak amplitude of the output voltage? Let $\alpha \simeq 1$.
Ans. $R_C = 8$ kΩ; $R_E = 8.6$ kΩ; 40 V/V; 0.4 V



阻抗匹配 $R_{sig} = R_{in} \approx r_e = 50\Omega$ ($R_E \gg r_e$ 并联忽略)

$$r_e = \frac{V_T}{I_E} \Rightarrow I_E = \frac{25mV}{50\Omega} = 0.5mA$$

$$I_C \approx I_E = 0.5mA \qquad g_m \approx \frac{1}{r_e} = 20mA/V$$

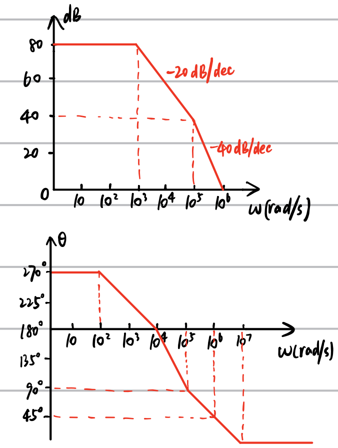
$$R_C = \frac{V_{CC}-V_C}{I_C} = \frac{5-1}{0.5mA} = 8k\Omega$$

$$R_E = \frac{V_E-V_{EE}}{I_E} = \frac{(0-0.7)-(-5)}{0.5mA} = 8.6k\Omega$$

$$G_v = \frac{R_{in}}{R_{sig}+R_{in}}A_v = \frac{R_{in}}{R_{sig}+R_{in}}g_m(R_C \parallel R_L) = \frac{1}{2}\times20\times4 = 40V/V$$

$$V_{out} = G_v \times v_{sig} = 10mV \times 40 = 0.4V$$

CH10-KP1-08：某放大器的传递函数 $T(jf) = \frac{-10^4}{(1+\frac{jf}{10^3})(1+\frac{jf}{10^5})(1+\frac{jf}{10^6})}$，分别画出其幅度和相位波特图（用折线近似法，并注意标明关键位置的坐标和折线斜率）。



CH10-KP2-02：当一阶RC无源低通滤波器输出信号的相移是 −45° 时，__A__。

A. 输出信号的幅度大约是输入信号的70 %。

B. 输出信号的幅度大约是输入信号的50 %。

C. 滤波器已经截止，输出信号的幅度几乎为0。

D. 随着信号频率的继续增加，输出信号相移逐渐逼近0°。

$\phi = -\arctan(2\pi fRC) = -\arctan(\frac{f}{f_c}) = -45° \Rightarrow f = f_c$ 截止频率

$$A_u = \frac{1}{\sqrt{H(\frac{f}{f_c})^2}} = \frac{1}{\sqrt{|H|^2}} = \frac{1}{\sqrt{2}}$$

$\boxed{\text{阻抗匹配}}$：前一级电路的R_o = 后一级电路的R_{in}

目的：
- ① 获得最大传输功率
  - $R_L \ll R_s$　R_L分不到电压
  - $R_L \gg R_s$　I很小
- ② 防止信号反射
  - 若电缆阻抗≠放大器输入阻抗，信号会反射

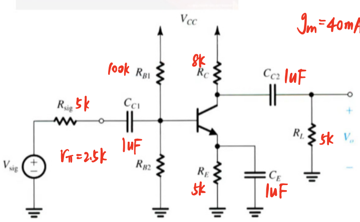
什么时候不用阻抗匹配？　电压放大（只传电压信号，不管电流）

Week 14

$R_B = R_{B1} \| R_{B2}$

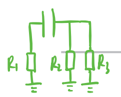
10.27 A common-emitter amplifier has $C_{C1} = C_E = C_{C2} = 1\ \mu F$, $R_B = 100\ k\Omega$, $R_{sig} = 5\ k\Omega$, $g_m = 40\ mA/V$, $r_\pi = 2.5\ k\Omega$, $R_E = 5\ k\Omega$, $R_C = 8\ k\Omega$, and $R_L = 5\ k\Omega$. Find the value of the time constant associated with each capacitor, and hence estimate the value of f_L. Also compute the frequency of the transmission zero introduced by C_E and comment on its effect on f_L.

Ans. $\tau_{C1} = 7.44$ ms; $\tau_{CE} = 0.071$ ms; $\tau_{C2} = 13$ ms; $f_L = 2.28$ kHz; $f_Z = 31.8$ Hz, which is much smaller than f_L and thus has a negligible effect on f_L.



$g_m = 40mA/V$ $\quad \frac{1}{g_m} = 25\Omega$

$r_\pi = \frac{\beta}{g_m} \Rightarrow \beta = 100$

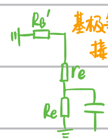


两侧：串
同侧：并

$R_1 + R_2 \| R_3$

$$\tau_{C1} = (R_{sig} + R_B \| r_\pi) C_{C1} = \left(5 + \frac{100 \times 2.5}{102.5}\right) \times 10^3 \times 1 \times 10^{-6} = 7.44\ ms$$

$$\tau_{CE} = \left[R_E \| \left(\frac{(R_{sig} \| R_B)}{\beta+1} + r_e\right)\right] C_{CE} = (5k \| (47+25)) \times 1 \times 10^{-6} = 0.071\ ms$$



基极等效来的接在 r_e 之上

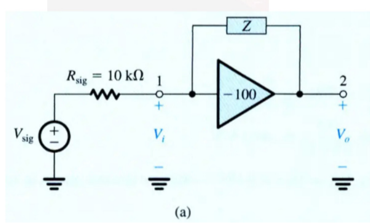
$$\tau_{C2} = (R_C + R_L) C_{C2} = 13\ ms$$

$$f_L = \frac{\omega_L}{2\pi} = \frac{1}{2\pi} \sum_{i=1}^{3} \frac{1}{\tau_i} = \frac{1}{2\pi}\left(\frac{1}{7.44} + \frac{1}{0.071} + \frac{1}{13}\right) \times 10^3 = 2.28\ kHz$$

$$\omega_Z = \frac{1}{C_E R_E} \qquad f_Z = \frac{\omega_Z}{2\pi} = \frac{1}{2\pi \times 10^{-6} \times 5 \times 10^3} = 31.8\ Hz$$

$\therefore f_Z \ll f_L$ 几乎无影响

Figure 10.15(a) shows an ideal voltage amplifier with a gain of -100 V/V and an impedance Z connected between its output and input terminals. Find the Miller equivalent circuit when Z is (a) a 1-MΩ resistance and (b) a 1-pF capacitance. In each case, use the equivalent circuit to determine V_o/V_{sig}.

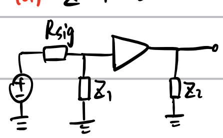


(a)

$Z_1 = \frac{Z_F}{1-A_v}$ $\quad \omega_{in} = \frac{1}{R_s(1+g_m R_D) C_F}$

$Z_2 = \frac{Z_F}{1-1/A_v}$ $\quad \omega_{out} = \frac{1}{R_D\left(1+\frac{1}{g_m R_D}\right) C_F}$

(a) $Z = 1M\Omega$

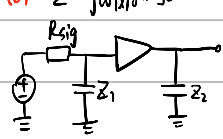


$$Z_1 = \frac{1 \times 10^6}{1+100} = 9.9 \times 10^3\ \Omega$$

$$Z_2 = \frac{1 \times 10^6}{1+\frac{1}{100}} = 0.999 \times 10^6\ \Omega$$

$$\frac{V_o}{V_{sig}} = \frac{Z_1}{R_{sig} + Z_1} A = -49.750\ V/V$$

(b) $Z = \frac{1}{j\omega 1 \times 10^{-12}}\ \Omega$



$$Z_1 = \frac{Z}{1-A} \Rightarrow \frac{1}{j\omega C_1} = \frac{1}{j\omega C_F(1-A)} \Rightarrow C_1 = C_F(1-A) = 101\ pF$$

$$Z_2 = \frac{Z}{1-1/A} \Rightarrow C_2 = C_F\left(1-\frac{1}{A}\right) \approx C_F = 1\ pF$$

$$\frac{V_o}{V_{sig}} = A \times \frac{1}{1 + j\omega C_1 R_{sig}}$$

$$\tau = R_{sig} C_1 = 10k\Omega \times 101\ pF = 1.01 \times 10^{-6}\ s$$

$$f_o = \frac{1}{2\pi\tau} \approx 157.6\ kHz$$

低通滤波 $\frac{V_o}{V_s} = \frac{k}{1 + j\frac{f}{f_o}}$

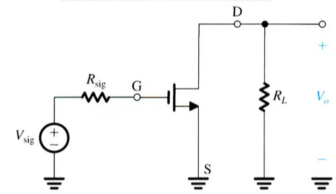
$$\therefore \frac{V_o}{V_{sig}} = \frac{-100}{1 + j\frac{f}{157.6kHz}}$$

10.6 For the CS amplifier specified in Example 10.1, find the values of A_M and f_H that result when the load resistance is reduced to 10 kΩ.

Ans. −13.3 V/V; 86.9 MHz

Example 10.1

Find the midband gain A_M and the upper 3-dB frequency f_H of an integrated-circuit CS amplifier fed with a signal source having an internal resistance $R_{sig} = 20\ k\Omega$. The amplifier has $R_L = 20\ k\Omega$, $g_m = 2\ mA/V$, $r_o = 20\ k\Omega$, $C_{gs} = 20\ fF$, and $C_{gd} = 5\ fF$. Also, find the frequency of the transmission zero.



$$A_M = -g_m (R_L \| r_o) = -13.33\ V/V$$

$$C_{in} = C_{gs} + (1 - A_M) C_{gd} = 91.65\ pF$$

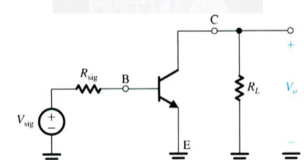
$$f_H = \frac{1}{2\pi C_{in} R_{sig}} = 86.87\ MHz$$

10.9 For the amplifier in Example 10.3, find the value of R_L that reduces the midband gain to half the value found. What value of f_H results? Note the trade-off between gain and bandwidth.

Ans. 2.44 kΩ; 923 kHz

Example 10.3

Find the midband gain and the upper 3-dB frequency of the common-emitter amplifier of Fig. 10.13(a) for the following case: $I_E = 1\ mA$, $R_{sig} = 5\ k\Omega$, $R_L = 5\ k\Omega$, $\beta_0 = 100$, $V_A = 100\ V$, $C_\mu = 1\ pF$, and $f_T = 800\ MHz$. Also, determine the frequency of the transmission zero.



$$g_m = \frac{I_C}{V_T} = \frac{1}{25}\ A/V \qquad r_\pi = \frac{\beta}{g_m} = 2.5\ k\Omega \qquad r_o = \frac{V_A}{I_C} = 100\ k\Omega$$

$$C_\pi + C_\mu = \frac{g_m}{\omega_T} = 8\ pF \qquad C_\mu = 1\ pF \Rightarrow C_\pi = 7\ pF$$

$$R_L' = R_L \| r_o = 4.76\ k\Omega$$

求新的 R_L^* 使 R_L' 减半 $\quad R_L^* \| r_o = 2.38\ k\Omega \Rightarrow R_L^* = 2.44\ k\Omega$

$$C_{in}^* = C_\pi + C_\mu (1 + g_m \cdot 2.38k\Omega) = 103.20\ pF$$

$$f_H^* = \frac{1}{2\pi C_{in}^* \cdot R_{sig}'} = 923.94\ kHz$$

Week 15

11.7 The circuit shown in Fig. E11.17 consists of a differential stage followed by an emitter follower, with series–shunt feedback supplied by the resistors R_1 and R_2. Assuming that the dc component of V_s is zero, and neglecting the base currents of the BJTs, show that the dc voltage at the output is approximately zero and find the dc operating current of each of the three transistors. Then find the values of A, β, $A_f \equiv V_o/V_s$, R_{in}, and R_{out}. Assume that the transistors have $\beta = 100$.

Ans. 0.5 mA, 0.5 mA, 5 mA; 85 V/V; 0.1 V/V; 8.95 V/V; 189.5 kΩ; 19.2 Ω.

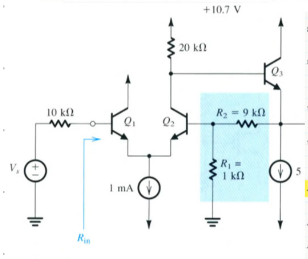


Figure E11.7

$g_{m1} := \frac{0.5\ mA}{25\ mV}$ $\quad g_{m2} := g_{m1}$ $\quad g_{m3} := \frac{5\ mA}{25\ mV}$ $\quad R_{11} := 0.9\ k\Omega$ $\quad R_{22} := 10\ k\Omega$

$$i_{e1}(v_s) := \frac{v_s}{\frac{R_S}{\beta+1} + \frac{R_{11}}{\beta+1} + \frac{\alpha}{g_{m1}} + \frac{\alpha}{g_{m2}}} \qquad R_{in3} := \left(\frac{R_{22} \cdot R_L}{R_{22}+R_L} + \frac{\alpha}{g_{m3}}\right) \cdot (\beta+1)$$

$$v_{b3}(v_s) := i_{e1}(v_s)\ \alpha \cdot \frac{R_C \cdot R_{in3}}{R_C + R_{in3}}$$

$$v_o(v_s) := \frac{v_{b3}(v_s)}{\frac{R_{22} \cdot R_L}{R_{22}+R_L} + \frac{\alpha}{g_{m3}}} \cdot \frac{R_{22} \cdot R_L}{R_{22}+R_L}$$

$A := v_o(1) = 85.305$ $\quad k := 0.1$ $\quad A_f := \frac{A}{1 + A \cdot k} = 8.951$

$$R_i := R_S + R_{11} + \left(\frac{\alpha}{g_{m1}} + \frac{\alpha}{g_{m2}}\right) \cdot (\beta+1) \qquad R_{o\_down} := \frac{R_{22} \cdot R_L}{R_{22}+R_L} \qquad R_{o\_up} := \frac{\alpha}{g_{m3}} + \frac{R_C}{\beta+1}$$

$$R_{in\_f} := R_i \cdot (1 + A \cdot k) \qquad R_o := \frac{R_{o\_down} \cdot R_{o\_up}}{R_{o\_down} + R_{o\_up}} \qquad R_{o\_f} := \frac{R_o}{1 + A \cdot k}$$

$R_{in} := R_{in\_f} - R_S = 189.188\ k\Omega$ $\quad R_{out} := \frac{R_L \cdot R_{o\_f}}{R_L - R_{o\_f}} = 19.167\ \Omega$

$I_{C1} = I_{C2} = \frac{1}{2} \times 1mA = 0.5mA$ (忽略I_B，基极开路，集电极电阻不影响，Q_1Q_2对称)

$I_{C3} = 5mA$ (忽略R_2的I_B，R_2分流很小)

$g_{m1} = g_{m2} = \frac{0.5mA}{25mV}$ $\quad g_{m3} = \frac{5mA}{25mV}$ $\quad R_{11} = R_1 \| R_2 = 0.9k\Omega$ $\quad R_{22} = R_1 + R_2 = 10k\Omega$

$$i_{e1}(V_s) = \frac{V_s}{\frac{R_s}{\beta+1} + \frac{R_{11}}{\beta+1} + \frac{\alpha}{g_{m1}} + \frac{\alpha}{g_{m2}}}$$

$V_{b3}(V_s) = i_{e1}(V_s)\ \alpha \cdot (R_C \| R_{in3})$ $\quad R_{in3} = \left[(R_{22} \| R_L) + \frac{\alpha}{g_{m3}}\right](\beta+1)$

$$V_o(V_s) = \frac{V_{b3}(V_s)}{(R_{22} \| R_L) + \frac{\alpha}{g_{m3}}} (R_{22} \| R_L)$$

$A = V_o(1) = 85.305$ $\quad k = \frac{R_1}{R_1 + R_2} = 0.1$ $\quad A_f = \frac{A}{1+AK} = 8.951$

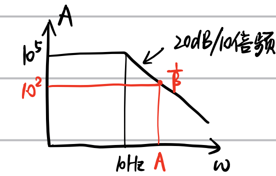
$R_i = R_s + R_{11} + \left(\frac{\alpha}{g_{m1}} + \frac{\alpha}{g_{m2}}\right)(\beta+1)$ $\quad R_o = \left(\frac{\alpha}{g_{m3}} + \frac{R_C}{\beta+1}\right) \| (R_{22} \| R_L)$

$R_{if} = (1+AK) R_i$ $\quad R_{of} = \frac{1}{1+AK} R_o$

$R_{in} = R_{if} - R_s = 189.88k\Omega$ $\quad R_{out} = \frac{R_L \cdot R_{of}}{R_L - R_{of}} = 19.167k\Omega$

11.21 Consider an op amp having a single-pole, open-loop response with $A_0 = 10^5$ and $f_P = 10$ Hz. Let the op amp be ideal otherwise (infinite input impedance, zero output impedance, etc.). If this amplifier is connected in the noninverting configuration with a nominal low-frequency, closed-loop gain of 100, find the frequency at which $|A\beta| = 1$. Also, find the phase margin.

Ans. 10^4 Hz; 90°



① $\frac{1}{\beta} = 100$ 为图红线

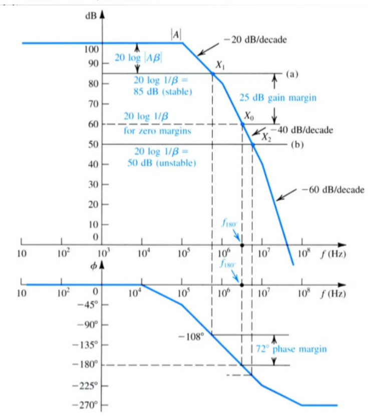
∴ A点频率为 $10Hz \times 10^3 = 10^4 Hz$

② 单极点，A点相位为 $-90°$

∴ $PM = 90°$

11.23 For the amplifier whose open-loop-gain frequency response is shown in Fig. 11.38, find the value of β that results in a phase margin of 45°. What is the corresponding closed-loop gain?

Ans. 10^{-4}; 80 dB



∵直线经过第二极点时 $PM = 45°$

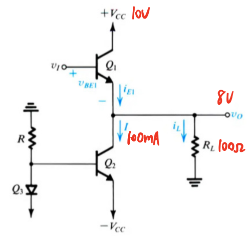
∴ $20\log_{10}\frac{1}{\beta} = 80$ $\quad \beta = 10^{-4}$

closed-loop-gain $= \frac{1}{\beta} = 80dB$

Week 16

12.3 For the emitter follower of Fig. 12.3, let $V_{CC} = 10$ V, $I = 100$ mA, and $R_L = 100\,\Omega$. If the output voltage is an 8-V-peak sinusoid, find: (a) the power delivered to the load; (b) the average power drawn from the supplies; (c) the power-conversion efficiency. Ignore the loss in Q_3 and R.

Ans. 0.32 W; 2 W; 16%



(1) $P_L = \frac{1}{2}\frac{\hat{V}_o^2}{R_L} = \frac{1}{2}\frac{8^2}{100} = 0.32W$

(2) $i_L = -80mA \sim 80mA$

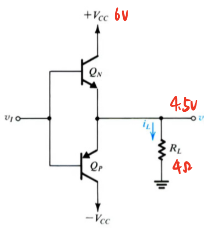
$i_{E1} = 20mA \sim 180mA$ 平均值 $\bar{I} = 100mA$

$P_s = 2V_{cc}\bar{I} = 2 \times 10 \times 100mA = 2W$

(3) $\eta = \frac{P_L}{P_s} = 16\%$

12.4 For the class B output stage of Fig. 12.6, let $V_{CC} = 6$ V and $R_L = 4\,\Omega$. If the output is a sinusoid with 4.5-V peak amplitude, find (a) the output power; (b) the average power drawn from each supply; (c) the power efficiency obtained at this output voltage; (d) the peak currents supplied by v_I, assuming that $\beta_N = \beta_P = 50$; and (e) the maximum power that each transistor must be capable of dissipating safely.

Ans. (a) 2.53 W; (b) 2.15 W; (c) 59%; (d) 22.1 mA; (e) 0.91 W



(1) $P_L = \frac{1}{2}\frac{\hat{V}_o^2}{R_L} = \frac{1}{2}\frac{(4.5)^2}{4} = 2.531W$

(2) 半周内平均 $\bar{I} = \frac{1}{2\pi}\int_0^{\pi}\frac{\hat{V}_o}{R_L}\sin\theta\, d\theta = \frac{\hat{V}_o}{\pi R_L}$

$P_{s+} = P_{s-} = \frac{1}{\pi}\frac{\hat{V}_o}{R_L}V_{cc} = \frac{1}{\pi} \times 1.125 \times 6 = 2.15W$

(3) $\eta = \frac{2.531}{2.15 \times 2} = 58.9\%$

(4) $\hat{I}_i = \frac{\hat{V}_o}{R_L}/(\beta+1) = \frac{1.125}{51} = 22.1mA$

(5) 最大功耗位于 $\hat{V}_o = \frac{2V_{cc}}{\pi}$ 处

单个晶体管功耗为 $P_{Dmax} = \frac{V_{cc}^2}{\pi^2 R_L} = 0.91W$